

# Storage – Data Storage Systems

## Data Platform

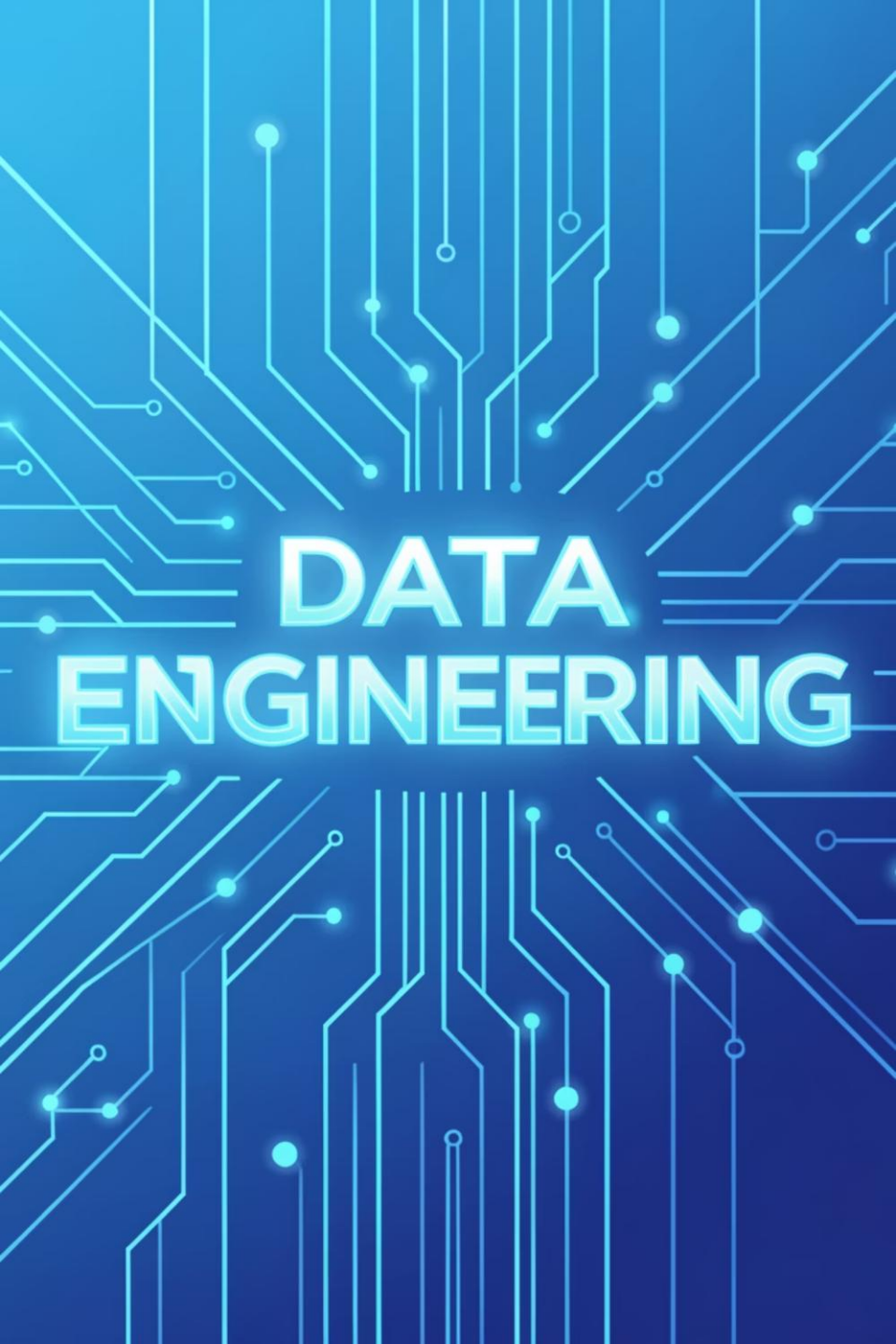
A **data platform** is ecosystem for integrated tools:

“All-in-one system” for:

- Storing data
- Processing data
- Analyzing data

**Example:**

BigQuery = **Storage + SQL + ML + Visualization** in same platform



# DATA ENGINEERING

## Data Engineering Spring '26 – DS463

Mar 30, 2026

Lecture 24

# Logistics: Your Course Essentials

## Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM  
AM – 12:20 PM

Friday (LH8): 11:00 AM – 11:50 AM

## Office Hours

Monday, Tuesday: 10:00 AM – 11:00 AM,  
AM, S18 2nd Floor NAB or by  
appointment.

## Credit Hours

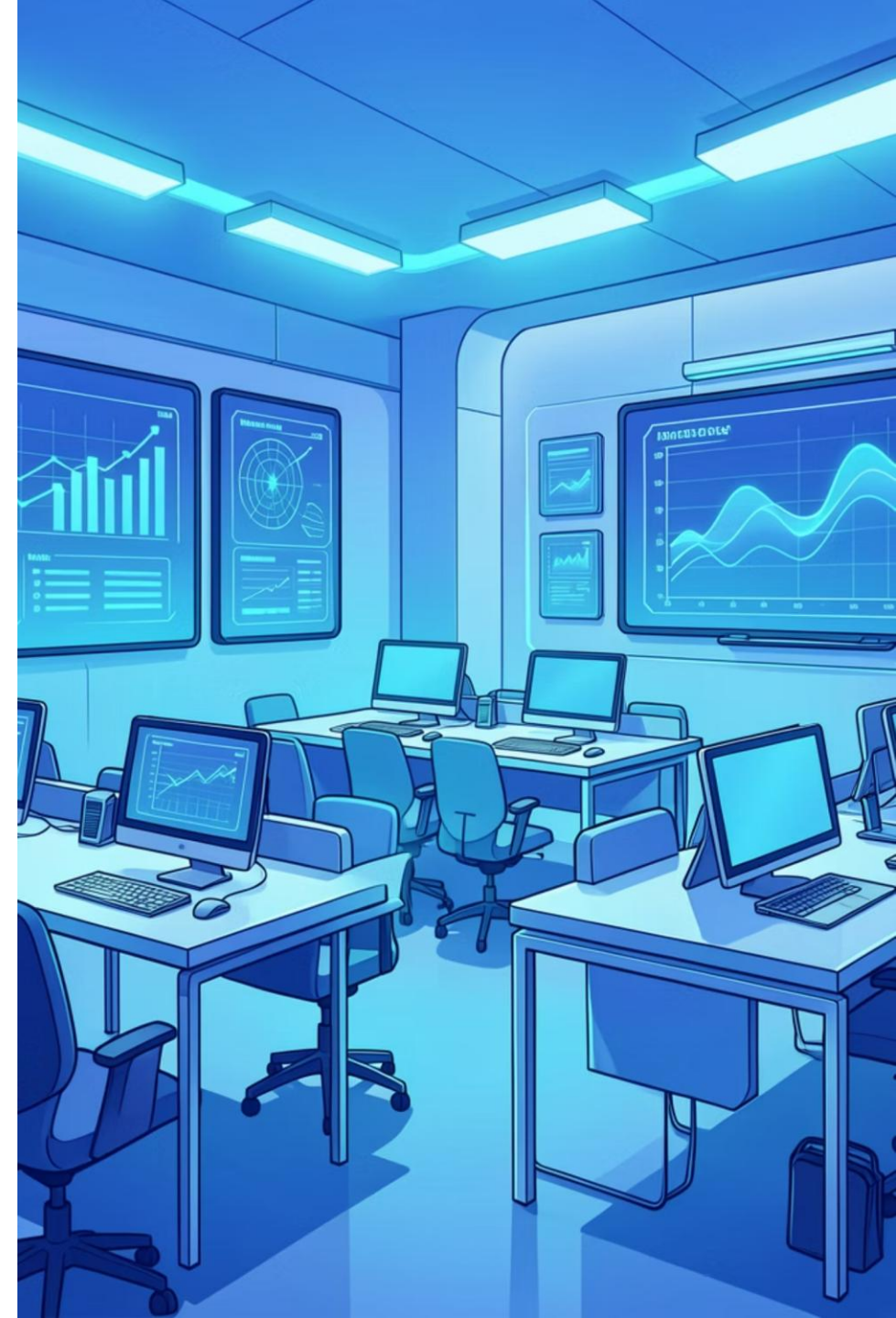
2 hours (Lectures) + 1 hour (Practical  
(Practical Session)

## Contact & Support

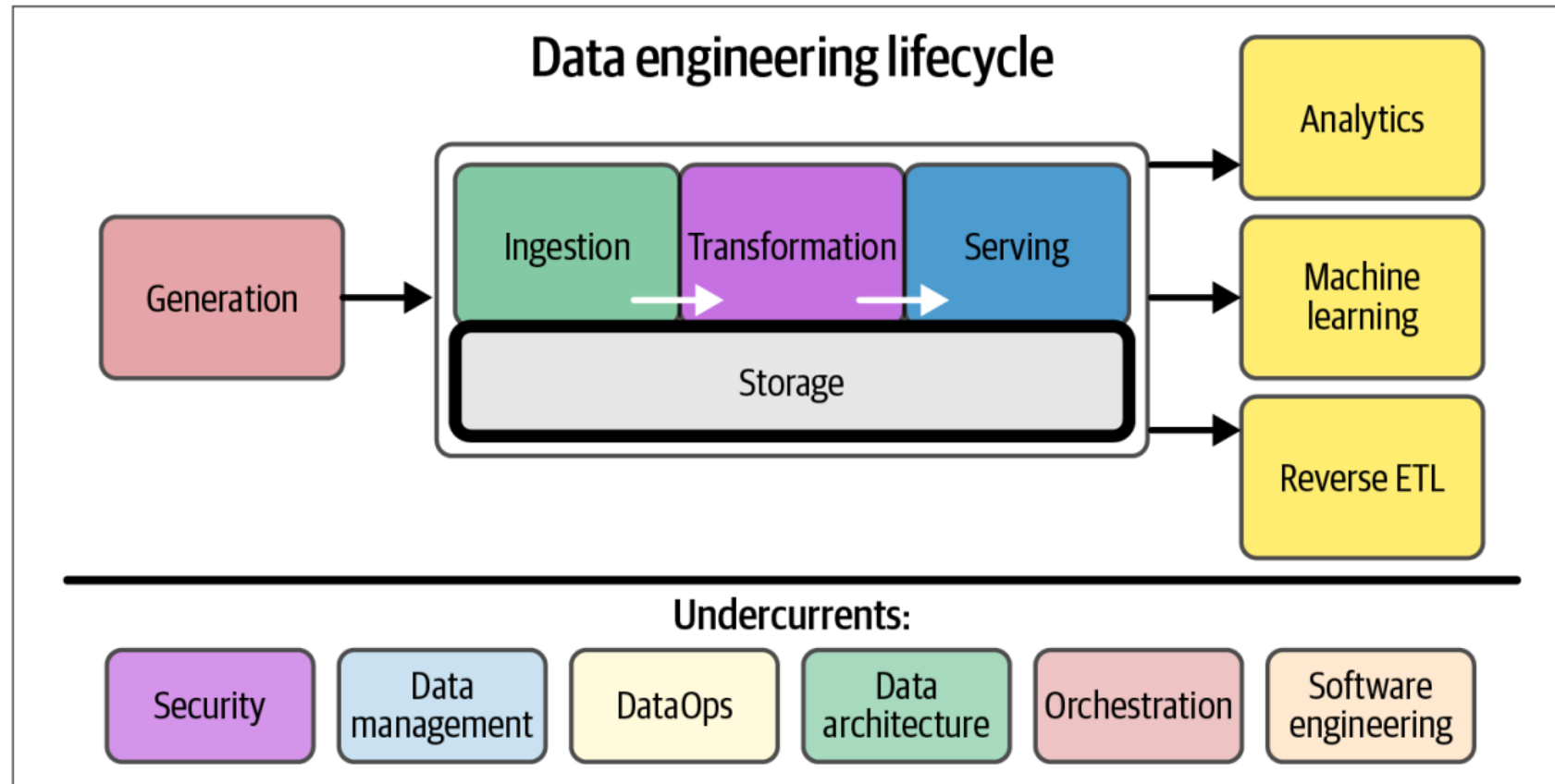
MS Teams: **w5t93yk**

Email: [farah.saeed@giki.edu.pk](mailto:farah.saeed@giki.edu.pk)

WhatsApp: Reach out for quick queries



# Data Engineering Lifecycle



*Figure 6-1. Storage plays a central role in the data engineering lifecycle*

**Storage** is the cornerstone of the data engineering lifecycle underlies its major stages—ingestion, transformation, and serving.

# Storage

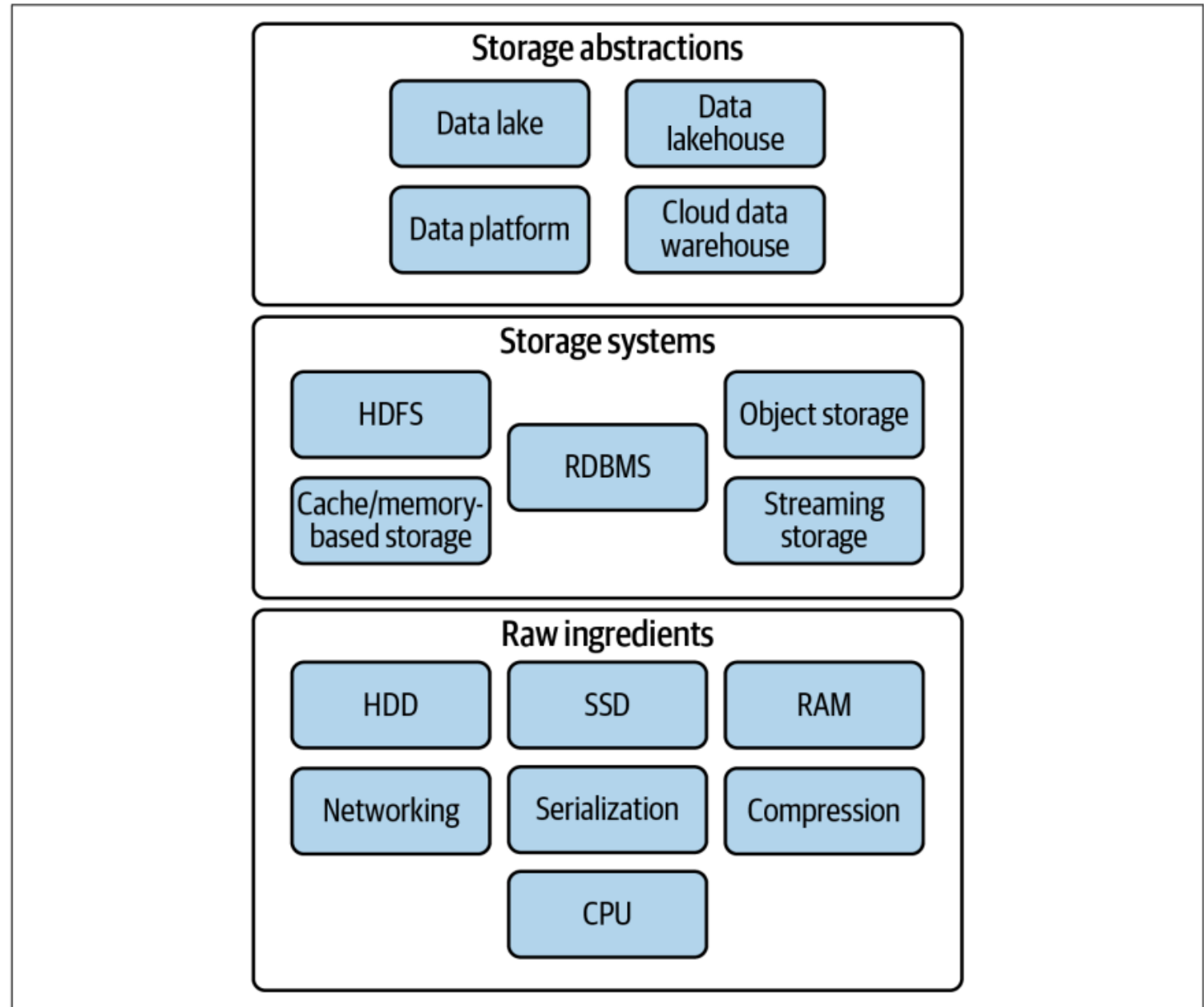


Figure 6-2. Raw ingredients, storage systems, and storage abstractions

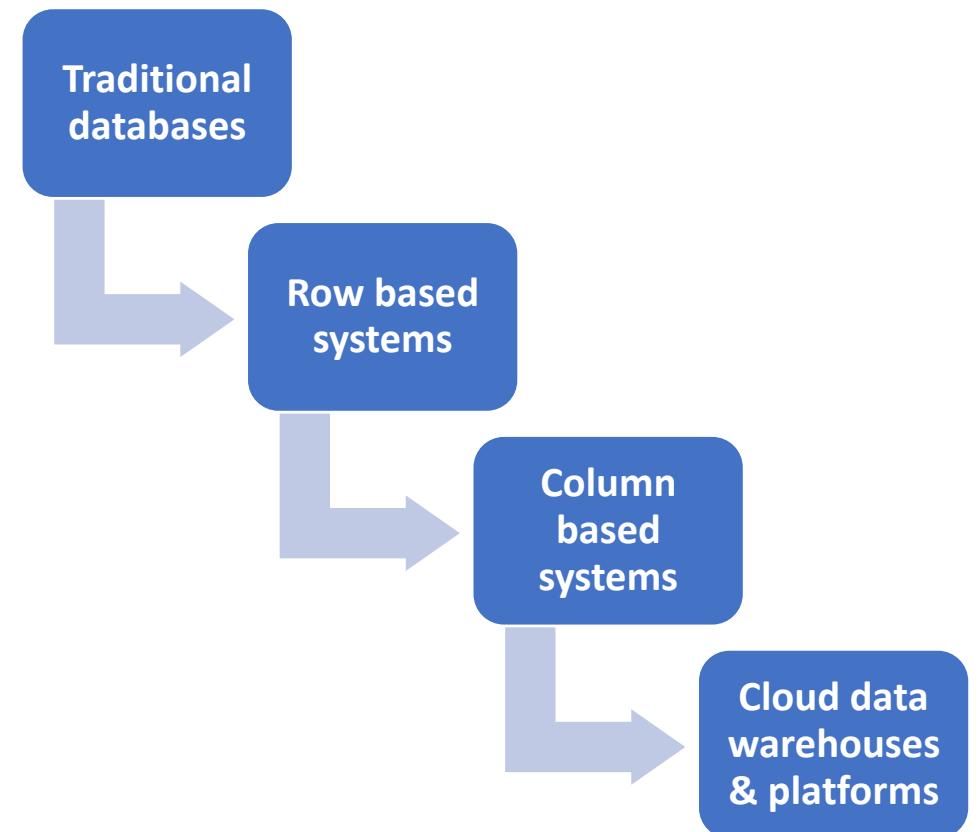
# Storage – Data Storage Systems

## Data Warehouse

### Data Warehouse

- A data warehouse is used to store and analyze large data.
- It is a standard system for OLAP (analytics & reporting)
- Centralized data architecture
- Organizational approach to managing data

### Evolution of Data Warehouse



# Storage – Data Storage Systems

## Data Lake

### Data Lake

- A data lake stores raw, unprocessed data
- Keeps data in its original format
- Used for storing very large amounts of data

### Early trends

- Built on **Apache Hadoop**
- Can store massive datasets
- Used for storing very large amounts of data

### Recent trends

#### Separation of Compute & Storage

- Storage and processing are separate systems
- Shift from Hadoop → cloud object storage
- More flexibility
- Lower cost

#### Need for More Structure

- Pure data lakes lacked:
- Schema (data organization)
- Update / delete capabilities

# Storage – Data Storage Systems

## Data Lakehouse

### Data Lakehouse combines:

- Data lake (raw storage)
- Data warehouse (structure & management)
- Stores data in object storage (like a data lake)
- Adds data warehouse features for better management

### Features

- **Structured tables & schemas**

Support for Updates, deletes, merges

- **Versioning (history & rollback)**

Can go back to older versions of data



- Not all data needs structure
  - Some can stay **raw/unstructured**
  - Some can be **organized into tables**
- Use structure **only when needed**

# Storage – Data Storage Systems

## Data Platform

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**Example:**

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# Storage – Data Storage Systems

## Stream to batch storage architecture

Combines

- Real-time (streaming) processing
- Batch (long-term) storage

Data is collected as a stream

Then sent to multiple systems (consumers)

### **Real-Time Consumers**

Process data instantly

Generate:

- Live statistics
- Dashboards

### **Batch Storage Consumer**

Stores data for

- Long-term use
- Batch analysis